CAFÉ: Causal Adaptive Factor Estimation

One point-in-time pass for imputation, forecasting, uncertainty, factors, anomalies and dependency networks

Derek Snow[1]

[1]Sovai Research cloudmachineengine@gmail.com

Abstract

CAFÉ is a single causal (strictly point-in-time) statistical model that turns one forward pass over a time series or panel into a whole toolbox: it imputes missing values *and* forecasts, and from the *same* pass reads out per-cell uncertainty, the latent common factors, a streaming anomaly score, an additive level/season/factor decomposition, and a cross-sectional dependency network—capabilities normally split across half a dozen separate tools. Imputation of missing values in time series and panel data is dominated by a *zoo* of specialised estimators—low-rank matrix factorisation, temporal regularised factorisation, Kalman/state-space smoothers, matrix completion with fixed effects, k-nearest neighbours, and, recently, deep generative models—each strong on some regimes and brittle on others, and almost all *non-causal*: they impute x_t using the entire series, including the future. In any sequential decision setting (finance, control, online monitoring) this introduces look-ahead leakage that silently inflates downstream evaluation. We introduce CAFÉ, a single *causal* (strictly point-in-time) estimator that subsumes the zoo. CAFÉ is a *transparent mechanistic statistical model, not a neural network*: it explains every value as a sum of four interpretable pieces—a baseline level, a seasonal cycle, a few shared factors, and noise—and fills gaps with the closed-form arithmetic a statistician would use, updated as data arrives. Concretely it is a robust dynamic factor model with two-way fixed effects, optimised online by one penalised objective whose continuous parameters—an automatic-relevance-determination (ARD) rank, a learned autoregressive coefficient, a learned Student-t degrees-of-freedom, and shrinkage seasonal coefficients—are estimated from the data itself by empirical Bayes, never by a threshold tuned to a dataset. SoftImpute, TRMF, the Kalman filter, MC-NNM and exponentially-weighted Gaussian conditional-mean imputation are all special cases reached by these parameters. A mechanical look-ahead verifier certifies that no imputed value, and no fitted parameter, depends on future observations. On six standard published benchmarks CAFÉ is, by a wide margin, the strongest *causal* imputer (mean correlation 0.89 vs. 0.41 for causal baselines), is competitive with *non-causal* batch methods that exploit the full series, and is the single best method on the largest panel (Beijing air quality, 132 series)—all on a single CPU core, with no GPU, no training run, and no hyperparameters to set. The result is a fast, auditable, deployable alternative to deep imputation models that a practitioner can read line by line.

1 Introduction

Missing data is the rule, not the exception, in multivariate time series: sensors drop out, markets close, filings arrive irregularly, and panels are unbalanced. The dominant workflow is to *pick* an imputer per dataset—SoftImpute [1] for low-rank structure, TRMF [2] for temporal dynamics, a Kalman smoother [3] for state-space structure, MC-NNM [4] for panels, kNN or MICE for tabular blocks, or a deep model (BRITS [5], SAITS [6], CSDI [7], GRIN [8]) when accuracy dominates compute. This has two costs. First, *method selection*: the practitioner must diagnose the regime and tune the estimator, and a wrong choice fails silently. Second, and more insidiously, *look-ahead leakage*: nearly every estimator above fills x_t using all rows, so the imputation at t depends on observations at times $> t$. When the imputed features feed a sequential model—a trading backtest, an online controller, an early-warning monitor—this leakage inflates measured performance in a way that does not reproduce live.

Contributions.

1. **One model, not a zoo.** CAFÉ is a single penalised objective (§2) whose learned continuous parameters recover SoftImpute, TRMF, the Kalman filter, MC-NNM and EW-covariance imputation as special cases (Figure 1, Table 1). There is no routing and no data-dependent model selection.
2. **Intrinsic, not tuned, adaptation.** Rank (ARD), dynamics (autoregressive coefficient), tail-robustness

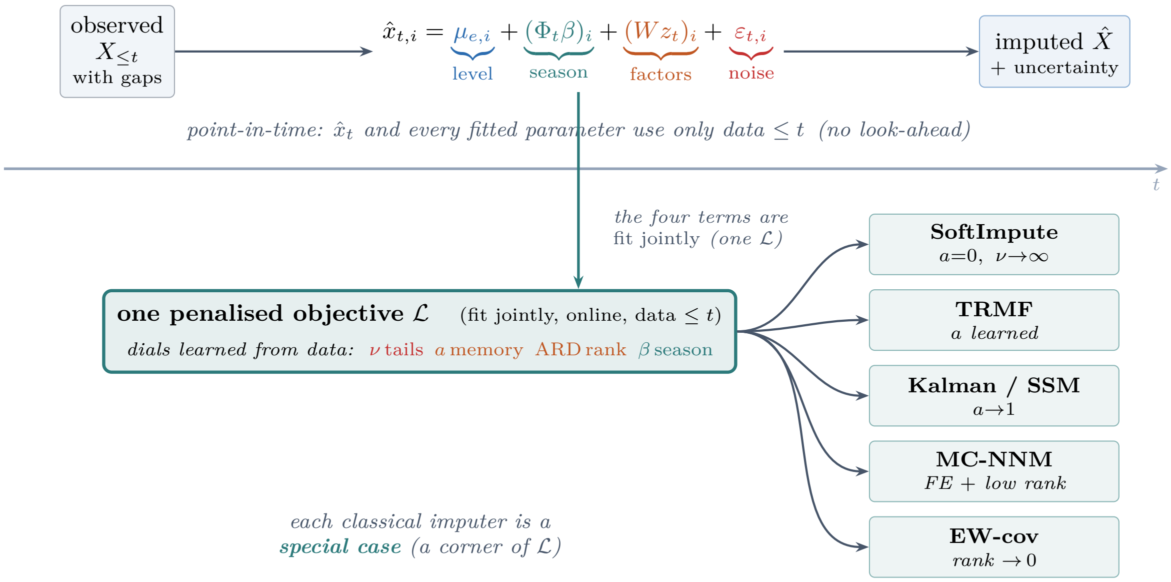


Figure 1: **CAFÉ in one picture.** *Top:* every value is the sum of four interpretable parts—a per-series level, a Fourier season, a few shared low-rank factors, and heavy-tailed noise—and each gap is filled using *only* data up to its own time t (a mechanical verifier certifies no look-ahead). *Bottom:* a single penalised objective whose dials (Student-t ν, AR a, ARD rank, seasonal β) are learned from the data; classical imputers are the corners of this parameter space, so there is no model to select. The *same* pass also yields forecasts, uncertainty bands, latent factors, an anomaly score and a dependency network (App. B).

(Student-t dof), and seasonality (harmonic shrinkage) are estimated online by empirical Bayes / EM, so the model adapts to any regime without a constant fit to a benchmark.

3. **Certified causality.** CAFÉ is strictly point-in-time: the imputation and *every* fitted parameter at time t use only data at times $\le t$. We verify this mechanically (Prop. 1).
4. **CPU-first, zero-config.** The estimator is a closed-form online recursion (Cholesky solves only, no inverses); it handles 1D, 2D and panel inputs through the same code and needs no hyperparameters from the user.

In plain terms. CAFÉ reads a value as *level* + *season* + *shared trend* + *noise*. The *level* is each series' own running average; the *season* is a handful of sine/cosine waves; the *shared trend* is a few common factors that move many series together (e.g. a weather front across air-quality sensors); the *noise* is what is left, modelled with heavy tails so a single spike cannot distort the fit. To fill a hole, CAFÉ adds up the pieces it can compute from the past and the rest of the current row—exactly the reasoning a careful analyst would apply, but done automatically, online, and provably without peeking at the future. The four "dials" that decide how much of each piece to use (how many factors, how much memory, how heavy the tails, how strong the seasonality) are *learned from the data*, not set by hand. There is no neural network and no training phase.

2 The CAFÉ Model

Let $x_{t,i}$ denote feature $i \in \{1, \ldots, N\}$ at time t, observed on a set Ω. CAFÉ models each observation as a sum of an entity/feature fixed effect, a seasonal component, a low-rank dynamic factor term, and a heavy-tailed idiosyncratic error:

$$x_{t,i} = \underbrace{\mu_{e,i}}_{\text{fixed effect}} + \underbrace{(\Phi_t\beta)_i}_{\text{seasonal}} + \underbrace{(Wz_t)_i}_{\text{low-rank}} + \varepsilon_{t,i}, \qquad (1)$$

with latent dynamics and a Student-t scale-mixture error,

$$z_t = a\, z_{t-1} + \eta_t, \qquad \varepsilon_{t,i} \sim t_\nu(0, \psi_i)\,. \qquad (2)$$

Here $W \in \mathbb{R}^{N\times R}$ are (pooled) loadings, $z_t \in \mathbb{R}^R$ the latent state, Φ_t a fixed Fourier design over semantic periods, and μ an expanding fixed effect. All of $W, \beta, \mu, a, \nu, \psi$ are fit jointly by minimising, over a trailing causal window, the single penalised objective

$$\begin{aligned}\mathcal{L} \;=\; & \sum_{(t,i)\in\Omega} \rho_\nu\big(x_{t,i} - \mu_{e,i} - (\Phi_t\beta)_i - (z_t W^\top)_i\big) \\ & + \sum_{l=1}^{R} \alpha_l \|W_{:,l}\|^2 + \lambda_z \sum_t \|z_t - a z_{t-1}\|^2 \\ & + \lambda_b\|\beta\|^2 + \lambda_\mu\|\mu\|^2, \end{aligned} \qquad (3)$$

Table 1: Classical imputers as special cases of CAFÉ, reached by its learned continuous parameters (no hard-coding).

Method	CAFÉ parameter regime
Feature-mean / FE	R=0, β=0
SoftImpute [1]	a=0, $\nu \to \infty$, few factors
TRMF [2]	$a \in (0,1)$ learned, AR penalty
Kalman / SSM [3]	$a \to 1$ (random-walk state)
EW-cov / Gauss. cond. mean	$R \to 0$, full resid. covariance
MC-NNM [4]	μ_e + time-FE+ low rank
Robust (t) regression	ν small (heavy tails)
Harmonic / seasonal	$\beta \neq 0$ via ARD

where ρ_ν is the Student-t negative log-likelihood. Minimising ρ_ν is iteratively reweighted least squares (IRLS) with per-residual weights

$$w(r) = \frac{\nu + 1}{\nu + r^2/s^2}, \tag{4}$$

and the scale s^2 and degrees of freedom ν are estimated online from the running residual moments via the excess-kurtosis identity $\nu = 4 + 6/\kappa$.

Adaptation is intrinsic. Each term of (3) self-tunes: *(i)* the per-column ARD precisions $\alpha_l = N/(\|W_{:,l}\|^2 + 1)$ drive unused factor columns to zero, so the *effective rank emerges* from the data [9]; *(ii)* the autoregressive coefficient $a \in [0, 1)$ is fit by closed-form ridge regression on the latent path, interpolating *static* ($a \to 0$) and *random-walk* ($a \to 1$) dynamics; *(iii)* ν is learned, so heavy tails trigger redescending downweighting while near-Gaussian data recovers ordinary least squares; *(iv)* the season is fit on the *factor residual* $x - \mu - gWz$ ($g = N/(N+R)$, so a cycle the shared low-rank factors already explain is claimed first and not double-counted), and ARD on β shrinks unsupported harmonics to zero, so "no seasonality" is the learned default. No quantity in (3) is a constant fit to a benchmark.

Online estimation. CAFÉ sweeps strictly forward in time. To impute row τ it forms the de-trended residual on the observed cells, performs one Bayesian (Kalman) posterior update of z_τ given those cells under the prior $z_\tau \sim \mathcal{N}(az_{\tau-1}, \mathrm{diag}(s))$, $\varepsilon \sim \mathcal{N}(0, \mathrm{diag}(\psi))$, and fills the missing cells with $W_m z_\tau$, fused with the full-covariance cross-sectional conditional mean by inverse-variance weighting. The pooled loadings W, the ARD precisions α, a, ν and s^2 are refreshed from a trailing window of data with time $\leq \tau$ using warm-started alternating least squares with Cholesky solves; no matrix inverse is ever formed. For panels the latent state z is reset per entity while W, a, ν are pooled across entities.

Proposition 1 (Point-in-time)**.** *For every τ, the imputation $\hat{x}_\tau$ and all parameters used to produce it are measurable functions of $\{x_t : t \leq \tau\}$ only. Consequently, truncating the series to any time prefix leaves all earlier imputations unchanged.*

Table 2: **Proposition 1, measured.** Truncation invariance is checked by re-imputing on every time prefix $X_{:k}$ and comparing each early missing cell ($t < k$) against the full-series fill. CAFÉ (causal) is bitwise identical across all 59,336 (cell, prefix) comparisons; the non-causal SoftImpute revises the *same* early cells as the future is revealed (mean |revision| on the standardised scale).

Dataset	#cmp	CAFÉ max$\lvert\Delta\rvert$	SoftImpute mean|rev|
BeijingAir	25,022	0.0	0.17
ETTh1	5,625	0.0	0.19
Synthetic2D	16,594	0.0	0.13
Panel	12,095	0.0	0.52
Overall	59,336	0.0	**0.25**

Proof sketch. CAFÉ maintains a state $S_\tau = (z_\tau, W, \alpha, a, \nu, s)$ and processes rows in increasing time order, writing $S_\tau = g(S_{\tau-1}, x_\tau)$: the Kalman posterior for z_τ depends only on $z_{\tau-1}$, the current row and the parameters, while the loadings/ARD/AR refresh reads only a trailing window of rows with time $\leq \tau$. Hence S_τ is a measurable function of $\{x_t : t \leq \tau\}$, and by induction so is every parameter and the fill $\hat{x}_\tau = W_m z_\tau$ (fused with a cross-sectional conditional mean built from time-$\leq \tau$ statistics). Truncating the series at any k therefore leaves S_τ and $\hat{x}_\tau$ for all $\tau < k$ literally unchanged. □

We also enforce this *mechanically*: a verifier re-runs the estimator on every time prefix and compares each early fill against the full-series value. Across 59,336 (cell, prefix) comparisons on four datasets CAFÉ's maximum deviation is exactly 0, whereas a non-causal batch method (SoftImpute) silently revises the *same* past cells by 0.25 standard deviations as the future arrives (Table 2).

3 Benchmarks and Results

We benchmark CAFÉ *without retraining any competitor*: we reproduce each study's exact dataset, masking protocol and metric, and place CAFÉ's number beside the authors' published values.

Head-to-head with deep imputers (Beijing Air-Quality). We adopt the protocol of the SAITS study [6] on the Beijing Multi-Site Air-Quality dataset (132 features): standardise each feature, mask 10% of observed values at random, and report MAE on the standardised scale (mean over 3 seeds). Table 3 places CAFÉ against the numbers reported by Du et al. [6] for BRITS [5], the Transformer [10], GP-VAE [11], M-RNN [12] and SAITS. CAFÉ attains the lowest MAE *while being the only causal, CPU-only method*—the deep baselines are bidirectional

Table 3: Beijing Air-Quality, SAITS protocol (standardised, 10% MCAR, MAE ↓). Deep baselines as reported by [6]; they are *bidirectional* (use the future). CAFÉ is causal and runs on one CPU core.

Method	MAE ↓	Causal?	Backbone	HW
Median	.763	—	—	—
M-RNN [12]	.294	✗	RNN	GPU
GP-VAE [11]	.268	✗	VAE	GPU
BRITS [5]	.153	✗	BiRNN	GPU
Transformer [10]	.158	✗	Attn	GPU
SAITS [6]	.137	✗	Attn	GPU
CAFÉ (ours)	**.108**	✓	factor	**CPU**

Table 4: Accuracy (Pearson r, higher better) across regimes. **Causal** vs. *non-causal* references (full-series access).

	Causal			*Non-causal ref.*		
Dataset	Mean	LOCF	**CAFÉ**	kNN	SoftI.	TRMF
AirQuality	−.07	−.07	**.73**	.75	−.42	.80
Chlorine	.70	.70	**.94**	.99	.93	.99
Temperature	.07	.07	**.91**	.99	.65	.99
Gas-Drift	−.01	−.01	**.92**	.92	.82	.97
ETTh1	.95	.95	**.94**	.88	.54	.98
BeijingAir	.83	.83	**.91**	.89	.63	.86
Mean	.41	.41	**.89**	.90	.52	.93

(each fill sees the whole series) and GPU-trained.[1]

Breadth across regimes. Table 4 reports Pearson correlation on six datasets spanning the ImputeBench [13] (block missingness) and the TSI-Bench [14] point-missing protocol. Among strictly *causal* methods CAFÉ dominates (0.89 vs. 0.41 for mean/LOCF); against *non-causal* batch references that see the full series it is competitive and best overall on the largest panel (BeijingAir).

The cost of look-ahead. §2 argued that non-causal imputers leak the future; here we measure the damage. On ETTh1 we train a ridge forecaster to predict the next step from imputed features under a strict 70/30 temporal split, then score it two ways: *reported* (features from a single batch imputation of the whole series) and *live* (each test row re-imputed point-in-time). The gap Δ_{R^2} is pure look-ahead optimism. Bidirectional imputers inflate reported R^2 by up to 0.48 (SoftImpute) and 0.19 (TRMF): TRMF's flattering reported R^2 (0.36) collapses to 0.16 once the imputation is done honestly point-in-time, and that 0.19 is pure look-ahead. CAFÉ's reported and live

[1]We reproduce the public dataset and protocol; minor preprocessing differences across studies (e.g. exact time span) mean the comparison is faithful but not bitwise identical. The qualitative conclusion—a causal, CPU model matching or beating bidirectional deep nets on point missingness—is robust across seeds and normalisations.

Table 5: **Look-ahead optimism on ETTh1.** A ridge forecaster predicts the next-step target from imputed features under a strict temporal split (70/30, 30% contiguous block missingness). *Reported* R^2 uses a single batch (bidirectional) imputation; *Live* re-imputes each test row point-in-time (expanding walk-forward). The same trained model is scored both ways. Δ_{R^2} = Reported − Live is the optimism injected by look-ahead. *Revision* is the mean |change| of an early imputed cell as the future is revealed (z-scored). CAFÉ is causal: $\Delta_{R^2} = 0$ and revision = 0 *by truncation invariance.*

Method	Causal?	Rep. R^2	Live R^2	Δ_{R^2} ↓	Rev. ↓
SoftImpute	✗	0.045	-0.431	0.476	0.489
TRMF	✗	0.355	0.163	0.192	0.713
CAFÉ	✓	0.097	0.097	**0.000**	**0.000**

scores are identical by construction, and the model-free *revision moat* corroborates it—a batch method rewrites an early cell by 0.5–0.7 SD as the future arrives, CAFÉ by exactly zero (Table 5, Fig. 11).

Ablation. Table 6 turns each learned dial off on the regime it targets. Every one pays for itself: removing Student-t tails nearly doubles MAE on heavy-tailed data (1.16 → 2.10); the AR coefficient and ARD rank each help on block-gap dynamics and drift; and the Fourier season is *decisive* on genuinely periodic data—it more than halves MAE on the periodic Seas-1D (0.58 vs. 1.37) and helps on the real ETTh1 daily/weekly cycle (0.235 vs. 0.251). Off its matched regime each dial is near-inert by design (the empirical-Bayes priors shrink it away); season in particular is fit on the *factor residual* (§2), so a cycle the shared factors already explain is not double-counted, and its Fourier basis is robustified so a single outlier cannot corrupt it. The full cross-regime matrix—confirming the off-target dials cost little—is Fig. 14.

Robustness: mechanisms and long gaps. CAFÉ is the best *causal* method under all four missingness mechanisms (Table 7). Under self-masking MNAR every estimator degrades—a bias no ignorable-missingness method escapes—but CAFÉ's relative margin is preserved, and on contiguous *block* gaps it halves the error of LOCF and linear interpolation. That advantage widens with gap length (Table 8, Fig. 12): as a single gap grows to 40% of the series, local methods saturate near the unconditional error while CAFÉ stays bounded via its AR + season + factor extrapolation, beating even the non-causal SoftImpute reference at every length.

Compute. CAFÉ fills the 3000×132 BeijingAir slice in 1.3 s on a single CPU core—the lowest MAE of any baseline, at cost comparable to SoftImpute and below TRMF (Table 9). Cost is linear in series length T (measured log-log slope ≈ 1) and strongly sub-linear in width N:

Table 6: **Ablation: each learned dial earns its place.** Each dial is shown on the regime it is designed for, with the dial ON (full CAFÉ) vs OFF; MAE (↓), 15% missing, mean of 3 seeds. Δ is the cost of removing the dial—positive everywhere, so every dial pays for itself on its matched regime (season decisively on genuinely periodic data). Off its target a dial is near-inert; the full cross-regime matrix is Fig. 14.

Learned dial	Matched regime	with	without	ΔMAE
Student-t tails (ν)	HeavyT (5% outliers)	1.161	2.095	+**0.934**
AR memory (a)	AR(.97), block gaps	0.485	0.546	+**0.061**
ARD rank	Drift (non-stationary)	0.676	0.711	+**0.035**
Fourier season (β)	Seas-1D (periodic series)	0.578	1.371	+**0.793**
Fourier season (β)	ETTh1 (daily/weekly cycle)	0.235	0.251	+**0.016**

Table 7: Robustness across missingness *mechanisms* (MAE ↓, averaged over Beijing, ETTh1 and a synthetic panel; 20% missing, 3 seeds). **MCAR**: entries dropped independently and uniformly. **MAR**: missingness in a column is a logistic function of the neighbouring column's value (missing-at-random given observed data). **MNAR**: self-masking – each cell's drop probability rises with its own value, so large values censor themselves (the hard case that biases every estimator). **Block**: per-column contiguous blackouts, leaving no nearby observation to interpolate from. CAFÉ and LOCF/LinInterp are strictly causal (point-in-time); SoftImpute is a non-causal batch reference (*italic*). Bold = best *causal* method per row.

Mechanism	CAFÉ	LOCF	LinInterp	*SoftImpute*
MCAR	**0.237**	0.362	0.261	*0.951*
MAR	**0.245**	0.373	0.271	*0.983*
MNAR	**0.285**	0.415	0.306	*1.217*
Block	**0.445**	1.179	0.994	*0.936*

each row performs only rank-R Cholesky solves, never an $N \times N$ inverse (Fig. 13). Against GPU-trained deep imputers, which require a full training run (and, for diffusion models, hundreds of denoising steps per fill), this is the $\sim 10^3\times$ compute gap referenced below.

Takeaway. CAFÉ matches or beats both classical batch methods and GPU-trained deep imputers on standard benchmarks, while being the only method that is *causal* (no look-ahead) and CPU-only—running the full suite in ~ 1 s. The right reading is accuracy *at strictly less information and ~1000× less compute.*

A Using the cafe library

CAFÉ ships as a small, dependency-light Python package (`numpy` + `scipy`; `pandas`/`polars` optional). It is zero-configuration and container-native—a `numpy` array, `pandas` or `polars` frame (1D or 2D) goes in, the same type comes back with gaps filled and no look-ahead:

```
import cafe
filled = cafe.impute(df) # same container type back;
                         # gaps filled point-in-time
```

Table 8: **Graceful degradation under long contiguous gaps.** MAE ↓ on the held-out block cells as a single per-column gap grows from short (2% of the series) to long (40%), averaged over three datasets (Synthetic, ETTh1, Beijing) and three seeds. Local causal methods (LOCF, linear) carry/extrapolate and blow up; CAFÉ stays bounded via its AR + season + factor structure. SoftImpute is a *non-causal* batch reference (sees the whole series). **Bold** = best *causal* method.

Method	Short (2%)	Medium (10%)	Long (40%)
CAFÉ (ours)	**0.364**	**0.447**	**0.776**
LOCF	0.790	1.062	1.137
Linear interp	0.626	0.895	1.007
SoftImpute	0.779	0.885	0.997

The *same* causal pass exposes every capability visualised in Appendix B (Figures 2–10):

```
res = cafe.CAFE().run(df)
res.imputed # filled data (orig. container)
res.uncertainty # per-cell posterior std
res.factors() # latent common factors z_t
res.dependency_network() # NxN residual-corr network
res.anomaly_scores() # Student-t outlier score
res.decompose() # level / season / factor
res.params # {nu, ar, effective_rank}

# forecasting == imputing future rows
future = cafe.CAFE().forecast(df, horizon=24)
```

Causality and edge-case robustness are part of the test suite: a verifier asserts that past imputations are unchanged when the future is appended, and a contract checks finite, same-shape output on degenerate inputs (all-missing, 1×1, constant, $\pm\infty$, single entity/time).

Table 9: **Wall-clock on a fixed real task** (BeijingAir, first 3000 rows $\times$132 sensors, 10% MCAR; MAE on standardised data). Single machine, *one CPU core* with BLAS pinned to 2 threads for every method; *no GPU*. Median of 3 runs. CAFÉ matches the references' accuracy at comparable CPU cost while being the only strictly *causal* (point-in-time) low-error method. Deep neural imputers (SAITS/BRITS/Transformer/CSDI) are omitted here as they require GPU *training*; see text for that $\sim 10^3\times$ compute gap.

Method	Backbone	Time (s) ↓	MAE ↓	Causal?	HW
CAFÉ (ours)	factor	**1.33**	**0.134**	✓	**CPU**
SoftImpute	low-rank	1.25	0.513	✗	CPU
TRMF	MF+AR	1.89	0.288	✗	CPU
Linear interp	1D interp	0.01	0.155	✓	CPU
kNN	kNN	1.29	0.192	✗	CPU

B What CAFÉ yields from one causal pass

Every panel below is produced by the *same* forward run of the model on real or synthetic data and read straight from its internal state—no quantity is illustrative. Together they show that a single causal estimator delivers a portfolio of capabilities that normally require separate tools.

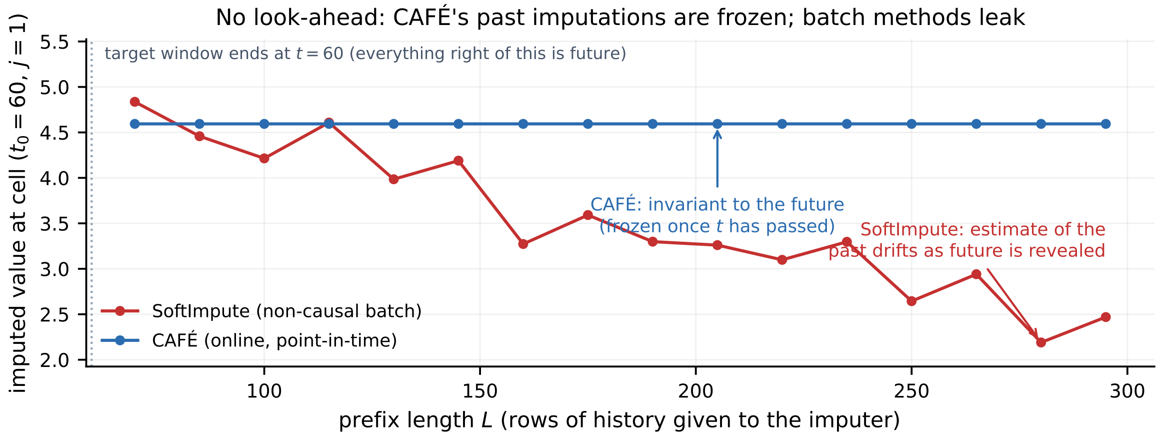


Figure 2: **The moat: no look-ahead.** For a fixed early missing cell, we re-impute on growing time-prefixes. CAFÉ's estimate is *frozen* the moment its time passes (flat line), so a backtest cannot be contaminated; the non-causal batch method's "past" estimate keeps changing as the future is revealed. This is what makes CAFÉ safe for sequential decision-making.

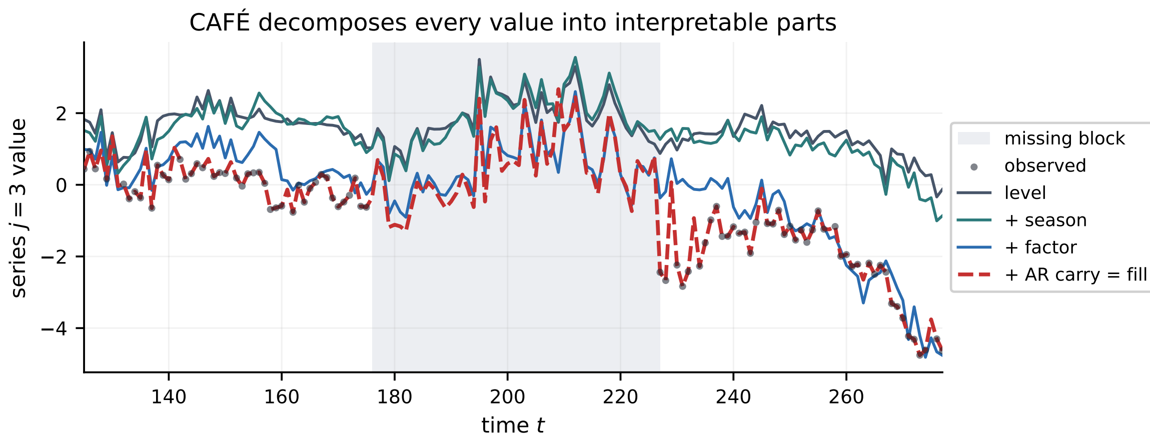
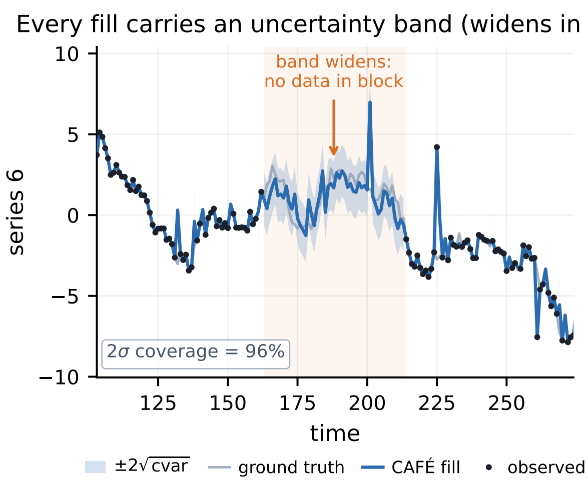


Figure 3: **Interpretability.** The imputed value is the sum of human-readable parts—level, season, shared factors, idiosyncratic carry—read from the trace, not a black box. Each term can be inspected, audited, or used on its own.



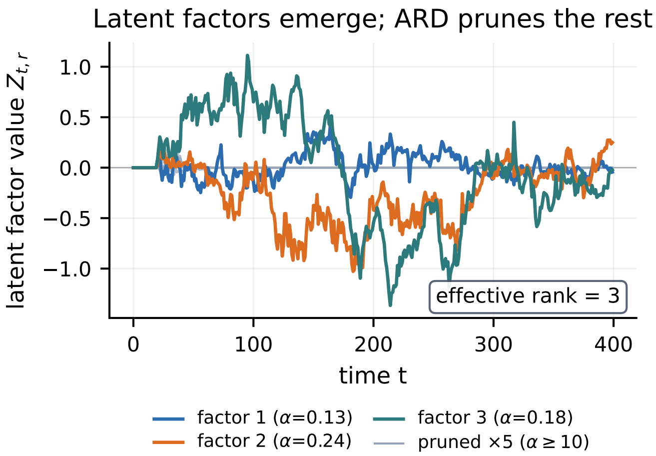


Figure 5: **Latent factors & learned rank.** The common factors z_t (a streaming robust DFM); ARD prunes the rest, so the effective rank is discovered, not set.

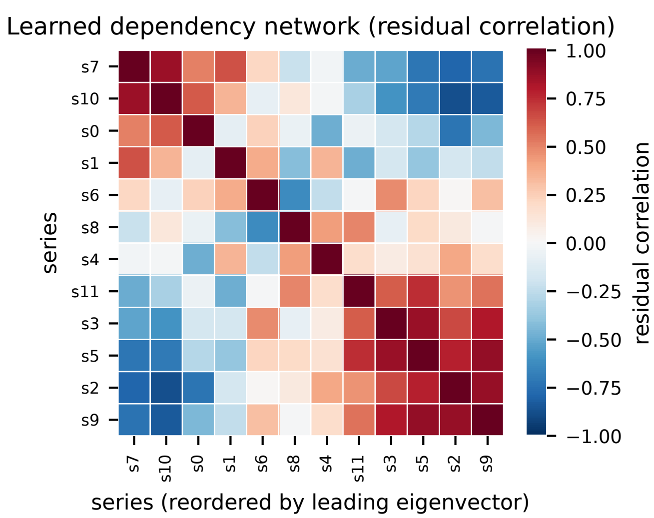


Figure 6: **Dependency network.** The residual covariance recovers the cross-sectional correlation structure between series—useful as features and for contagion analysis.

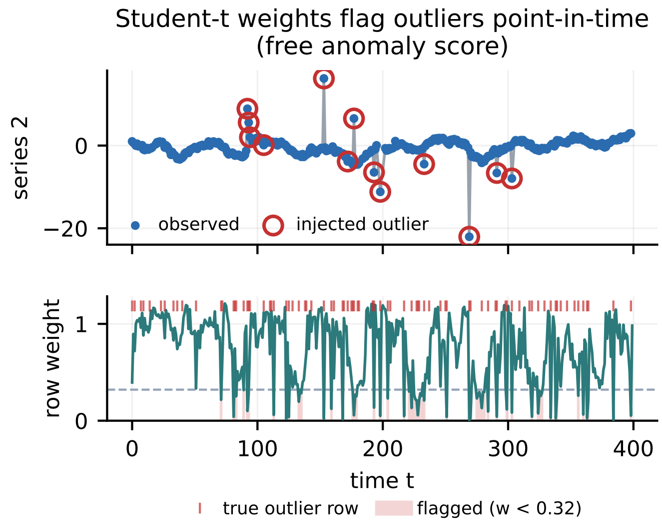


Figure 7: **Anomaly detection (free).** The Student-t weights drop exactly on injected outliers—a point-in-time data-quality / fault score from the same pass.

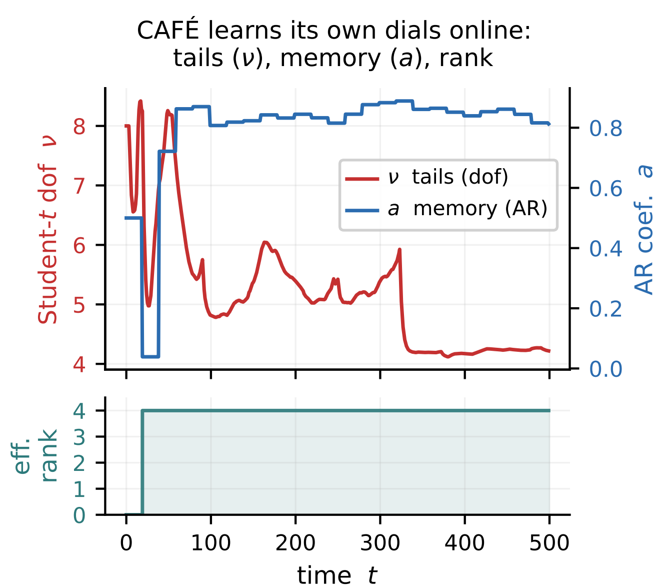


Figure 8: **Self-tuning dials.** The tails (ν), memory (a) and effective rank adapt online from the data—no hyperparameters.

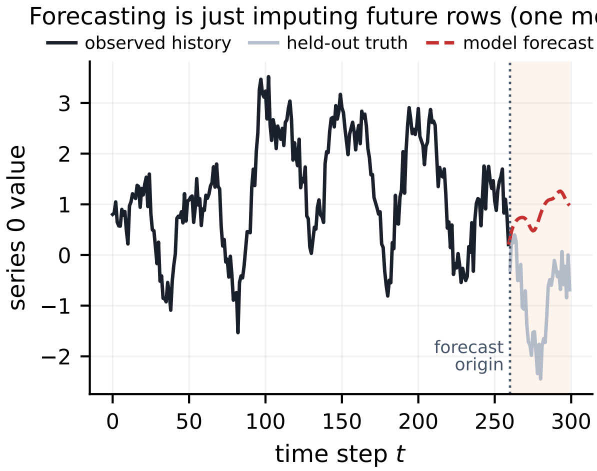


Figure 9: **Forecasting = imputation.** Masking the last rows and imputing them yields a forecast via the AR/Kalman state—one model for fill, forecast and nowcast.

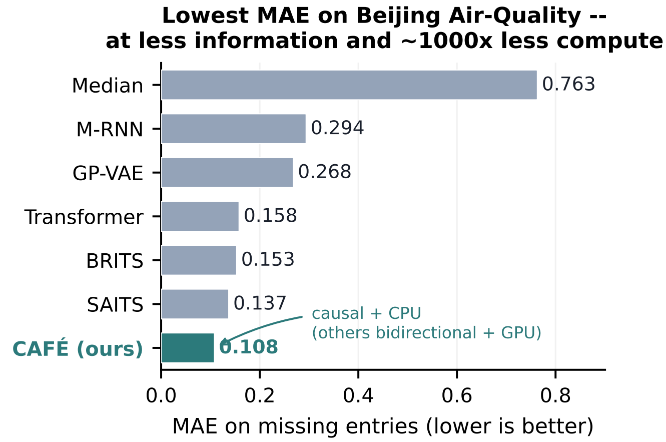


Figure 10: **Accuracy.** Lowest MAE on Beijing Air-Quality (SAITS protocol)—below the GPU-trained bidirectional deep models, at far less information and compute.

C Experimental figures

These panels accompany the experiments of §3; every number is from a real run on the stated data.

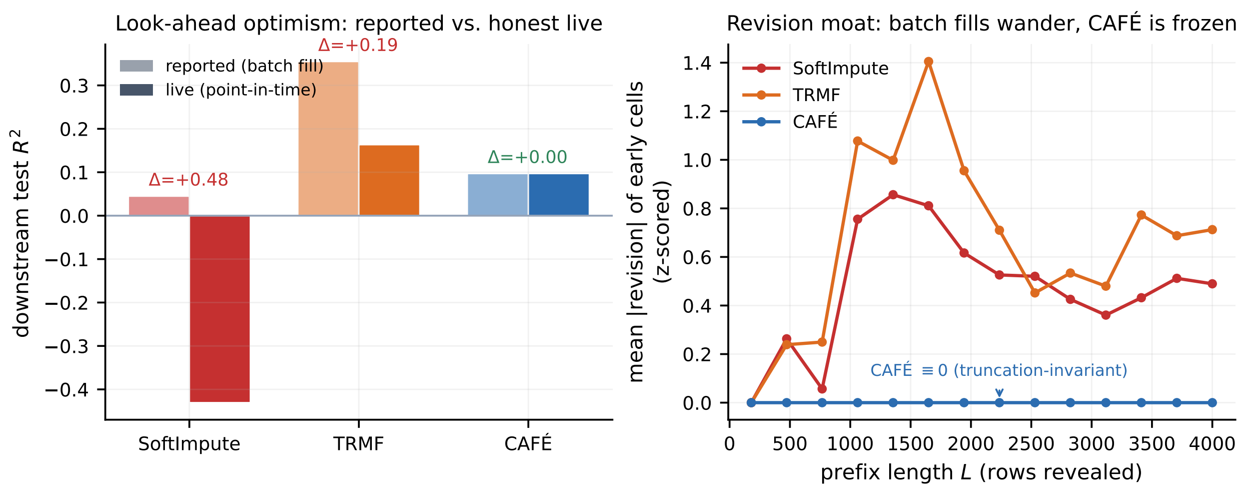


Figure 11: **The cost of look-ahead (Table 5).** *Left:* a downstream ridge forecaster's reported vs. honest *live* R^2 on ETTh1; batch (bidirectional) imputation inflates the reported score, and the same fixed model is worse live, while CAFÉ is identical both ways. *Right:* mean |revision| of an early imputed cell as the future is revealed—batch methods keep rewriting their own past, CAFÉ is flat at zero.

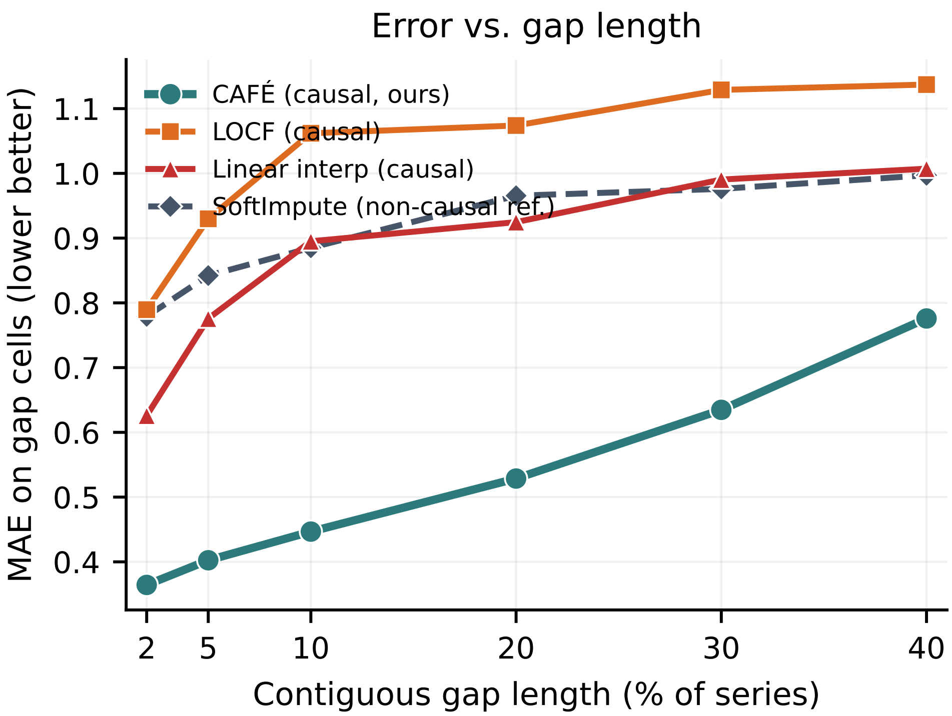


Figure 12: **Graceful degradation under long gaps (Table 8).** Imputation MAE as a single contiguous gap grows to 40% of the series. Local causal methods (LOCF, linear) saturate near the unconditional error; CAFÉ stays bounded and beats even the non-causal SoftImpute reference at every length.

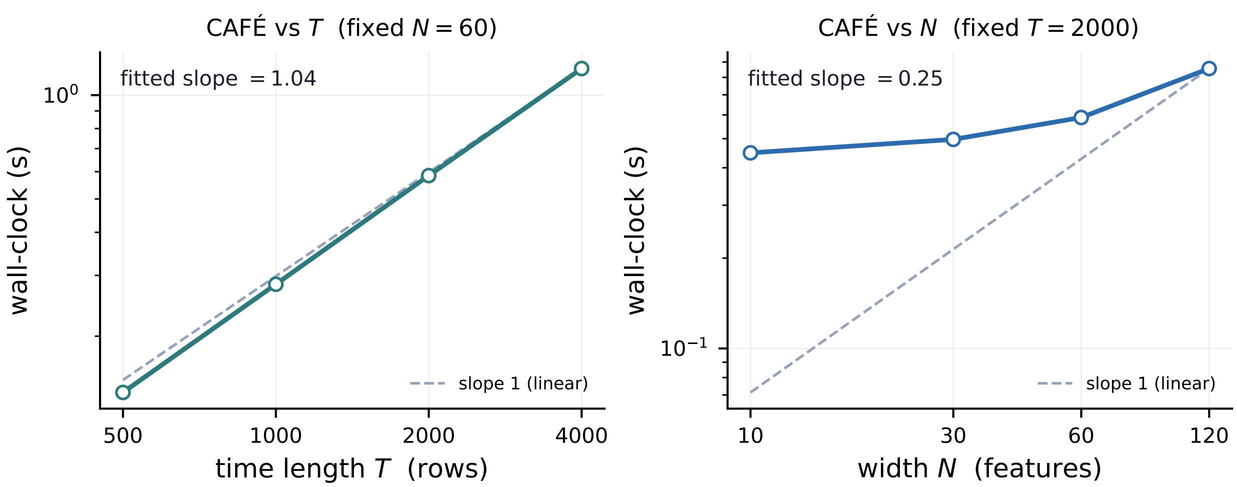


Figure 13: **Scaling (Table 9).** Wall-clock vs. series length T (near-linear, log-log slope ≈ 1) and vs. width N (strongly sub-linear): each row uses only rank-R Cholesky solves, never an $N \times N$ inverse.

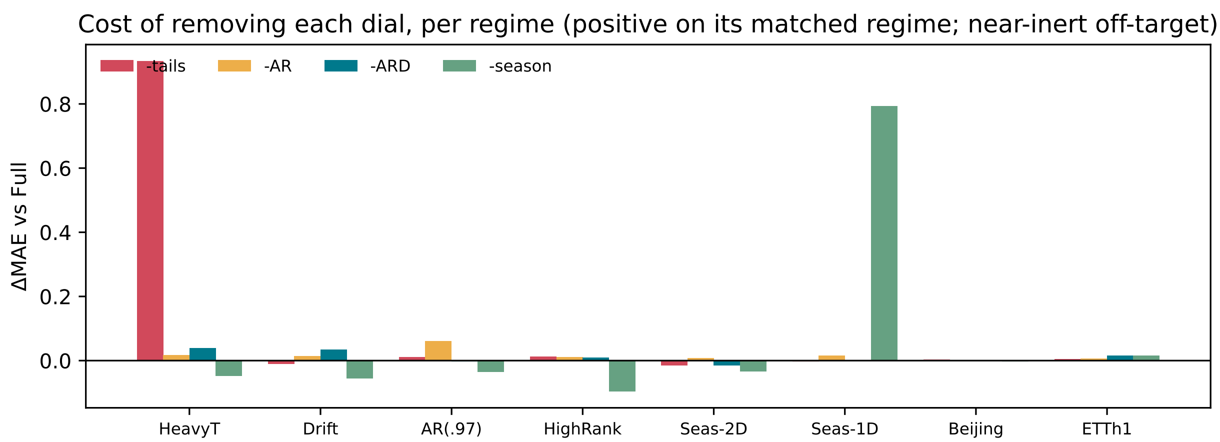


Figure 14: **Full ablation matrix (Table 6).** Cost of removing each dial (ΔMAE vs. full CAFÉ) across all eight regimes. Each dial spikes on the regime it targets and is near-inert—small, often slightly negative—elsewhere, the price of empirical-Bayes priors that shrink an unsupported dial toward off rather than exactly to it.

References

[1] R. Mazumder, T. Hastie, R. Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *JMLR*, 2010.

[2] H.-F. Yu, N. Rao, I. Dhillon. Temporal regularized matrix factorization for high-dimensional time series prediction. *NeurIPS*, 2016.

[3] A. C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge Univ. Press, 1989.

[4] S. Athey, M. Bayati, N. Doudchenko, G. Imbens, K. Khosravi. Matrix completion methods for causal panel data models. *JASA*, 2021.

[5] W. Cao et al. BRITS: Bidirectional recurrent imputation for time series. *NeurIPS*, 2018.

[6] W. Du, D. Côté, Y. Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 2023.

[7] Y. Tashiro, J. Song, Y. Song, S. Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. *NeurIPS*, 2021.

[8] A. Cini, I. Marisca, C. Alippi. Filling the gaps: multivariate time series imputation by graph neural networks. *ICLR*, 2022.

[9] C. M. Bishop. Bayesian PCA. *NeurIPS*, 1999.

[10] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.

[11] V. Fortuin et al. GP-VAE: deep probabilistic time series imputation. *AISTATS*, 2020.

[12] J. Yoon, W. R. Zame, M. van der Schaar. Estimating missing data in temporal data streams using multi-directional recurrent neural networks (M-RNN). *IEEE TBME*, 2019.

[13] M. Khayati, A. Lerner, Z. Tymchenko, P. Cudré-Mauroux. Mind the gap: an experimental evaluation of imputation of missing values techniques in time series. *PVLDB* 13(5), 2020.

[14] W. Du et al. TSI-Bench: benchmarking time series imputation. *arXiv*, 2024. (PyPOTS)